

Operations Research for Health Care

Lesson 3/15

João Flávio de Freitas Almeida

LEPOINT: Laboratório de Estudos em Planejamento de Operações Integradas
Departamento de Engenharia de Produção

Universidade Federal de Minas Gerais
Escola de Engenharia

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Outline of this lecture

Health care facility location

Non-emergency health care location

Emergency health care location

Discrete location problems

Health care location-allocation modelling

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The health care industry [Ahmadi-Javid et al., 2017]

Healthcare industry: Integration of various sectors to provide healthcare products and services. The major players are:

- ▶ **Producers:** all health suppliers, such as pharmaceutical companies, medical-surgical product companies, and medical equipment manufacturers;

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- ▶ **Fiscal intermediaries:** Insurance agencies and health maintenance organizations.

The healthcare facility location [Ahmadi-Javid et al., 2017]

The location of healthcare facilities is one of the most important strategic issues in healthcare systems.

- ▶ This is because **poorly located** facilities or an **improper number of facilities** can greatly *increase capital and inventory costs and degrade customer services*;

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- ▶ Researchers classified studies into location categories. One of them is measuring the effectiveness of **past location decisions** [Rahman and Smith, 2000];

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- ▶ Researchers classified studies into location categories. One of them is measuring the effectiveness of **past location decisions** [Rahman and Smith, 2000];
- ▶ **Accessibility models** are extensions of location models whose goals were predominantly to maximize coverage or to minimize average distance.

From this perspective, facility location modelling for healthcare is more critical than similar modelling for other areas.

The healthcare facility location [Ahmadi-Javid et al., 2017]

Primary, secondary, and tertiary care facilities (basic care units, clinics, hospitals).

- ▶ **Primary care** is usually performed by a general practitioner and has *great potential for referral to specialty services*, which are secondary, tertiary, or quaternary care; or non-medical services.
- ▶ **Primary health care** services include health promotion and disease prevention, and population-level public health functions.

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- ▶ **Primary health care** services include health promotion and disease prevention, and population-level public health functions.
- ▶ It should be mentioned that hospitals and most clinics provide, *beside primary care*, speciality services, which are **secondary, or tertiary care**. They can be classified into different levels, making them **hierarchical location problems**.

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Regarding location management, healthcare facilities are classified into non-emergency and emergency.

Non-emergency health care location:

- ▶ **Facilities:** Primary, secondary and tertiary care, basic-care units, clinics, hospitals (discrete location models, and hierarchical);

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Non-emergency health care location:

- ▶ **Facilities:** Primary, secondary and tertiary care, **basic-care units, clinics, hospitals** (discrete location models, and hierarchical);
- ▶ **Blood banks:** The use of blood and blood products on a daily basis (inventory management, and supply chain management). Different types of blood banks (e.g., blood transfusion providers, blood centres, blood stations and mobile units) are highly important in health systems. Human blood is a **perishable, scarce**, and valuable product with a lifetime of only 21 days. Moreover, both **demand and supply of blood are stochastic**.

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Non-emergency health care location:

- ▶ **Organ transplant:** (location): The demand for organs continues to exceed the supply: long waiting lists; combination of donor and recipient; **time** that elapses between donor notification and transplantation **is vital**;

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- ▶ **Medical laboratories:** Services in testing and diagnosing infection and contagious diseases. (location) consider **laboratory capacities, budgets, and maximum transportation times** (between origins and laboratories).

The healthcare facility location [Ahmadi-Javid et al., 2017]

Non-emergency health care location:

- ▶ **Mobile HC units:** where there is no fixed HCF (i.e., hospital, clinic or BCU). Location of mobile HCFs consider a **closed tour with selected stops**;

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- ▶ **Drugstores:** Most units are **private** and easy to construct. To reach patients, set the **optimal number** and proper **location** of them in increasing system **profitability** (by cost reduction and revenue increase);

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- ▶ **Long-terms centres:** Aging of the population. Provide medical (including nursing care) and social services to inpatients, and concurrent determination of **optimal capacity levels (number of beds), inventory levels, and locations**.

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Efficiency, effectiveness, and equity are major criteria to evaluate accessibility to non-emergency HCFs. Cost minimization is another major objective that is used in HCF location problem. A major policy issue for a national health system in particular is the efficiency and effectiveness of the public HCFs concerning their locations in local communities. It should be noted that cost minimization has recently gained particular attention for these PCFs.

- ▶ There are many links between the different levels of Primary Care Facilities (PCFs) which makes *it impossible to solve the location problems for each level independently*[Hodgson, 1988]. However, hierarchical location models have not received much attention.

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Emergency health care location (permanent and temporary):

- ▶ **Emergency centres** (permanent): Provide medical care to both walk-ins (unscheduled patients) and emergency patients transported by an ambulance, who require immediate and urgent medical treatment. **Priority queuing-based covering location models** (hospitals and ambulance stations);

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- ▶ **Trauma centres** (permanent): Hospitals that provide specialized medical and nursing care to patients with traumatic injuries, and typically have **helicopter** platforms to transfer patients. Although trauma patients require immediate transfer to trauma centres, the establishment of helicopter platforms and using air transport are **expensive**;

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- ▶ **Ambulance stations** (Permanent): EMSs (emergency medical services) play a pivotal role in health systems, which are generally concerned with providing out-of-hospital acute medical care and **transferring** patients to emergency centres, emergency departments within hospitals, or trauma centres for definitive care;

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Emergency health care location (permanent and temporary):

- ▶ **Disasters:** (temporary) Temporary medical centres (TMCs) are providers of healthcare services to people who are affected by disasters or **large-scale emergencies**, which may be of a catastrophic nature. This class contains **field hospitals**, Red Crescent and/or Red Cross **tents**;

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- ▶ **Points of dispensing** (temporary): A mass medication dispensing site that is capable of providing medicine and medical supplies (e.g., vaccines, drugs, and therapeutics) to protect the general population in infectious disease disasters (e.g., epidemics, **pandemics**, or an outbreak of an emerging infectious disease).

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Overview of discrete location problems [Ahmadi-Javid et al., 2017]

- ▶ **Taxonomy** for location problems based on the space: **Continuous** (in which facilities may be located anywhere in the feasible region) or **Discrete** (in which they can be established only at **candidate locations** that can include the demand points);
- ▶ **Extentions**: Stochastic location problems, hierarchical location problems, multi-criteria location problems, hub location problems, dynamic and online location problems, competitive location problems;

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- ▶ **Classification** of discrete location problems [Daskin, 2008]: **covering**-based problems, **median**-based problems, and other problems.

Overview of discrete location problems [Ahmadi-Javid et al., 2017]

- ▶ **On the one hand**, covering-based and median-based problems are well-known facility location problems for **modelling real-world situations**.

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- ▶ **On the one hand**, covering-based and median-based problems are well-known facility location problems for **modelling real-world situations**.
- ▶ **On the other hand**, important problems, such as p-center and p-median location problems on a general network or in the plan **are NP-hard** for $p > 1$. These are problems with *no known polynomial-time exact solution algorithms*. It means that the time required to solve an instance of these problems exactly may increase very rapidly as the size of the problem instance grows, often well beyond any reasonable timeframe.

Covering-based problems [Ahmadi-Javid et al., 2017]

For determining the location of **emergency service facilities**: Demand locations are within a specific coverage distance (or time) from supply facilities.

- ▶ **Set covering**: Minimizing the number of established facilities or the total location cost, given a specified level of *demand coverage that must be achieved*. Find the number and location of facilities such that all demand points are within a specified travel distance (or time) of the facilities that serve them;

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- ▶ **Maximal covering**: Determine the location of p facilities in order to maximize the demand covered within a *pre-specified maximum coverage distance*, taking into account the level of demand at each point;

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- ▶ **Maximal covering**: Determine the location of p facilities in order to maximize the demand covered within a *pre-specified maximum coverage distance*, taking into account the level of demand at each point;
- ▶ **P-center**: *Minimize the maximum travel distance (or time) among all demand points and the allocated facilities, considering that every demand point is covered*. When the facilities are uncapacitated, the demand points are assigned to the closet open facilities. Also referred to as **location-allocation** problems since they require simultaneous *facility location and allocation of the demand points to the open facilities*. P-center *find the smallest possible coverage distance such that every node is covered*.

Median-based problems [Ahmadi-Javid et al., 2017]

Locate facilities at candidate points so as to *minimize the weighted average distance costs between demand points and destination* facilities. These locations are the **medians of the network**.

This class of problems may be referred to as **location-allocation** problems as they determine both location and allocation decisions.

- ▶ **P-median**: These problems aim to locate p facilities in a network;

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- ▶ **P-median**: These problems aim to locate p facilities in a network;
- ▶ **Fixed charge**: Closely related to P-median. However, it attempts to *minimize the total cost of facility opening and travelling*.

Overview of discrete location problems [Ahmadi-Javid et al., 2017]

Discrete location problems [Daskin and Dean, 2004, Daskin, 2008]:

- ▶ **Covering-based problems:**

- ▶ Set-covering
- ▶ Maximal covering
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Discrete location problems [Daskin and Dean, 2004, Daskin, 2008]:

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- ▶ Maximal covering
- ▶ P-center

- ▶ **Median-based problems:**

- ▶ P-median
- ▶ Fixed-charge

- ▶ **Extension:**

- ▶ Health care facility location

Set-covering [Ahmadi-Javid et al., 2017]

I : The set of demand points.

J : The set of candidate locations for Health Care Unit (HCU)

- ▶ d_{ij} The travel time from demand point $i \in I$ to candidate location $j \in J$.
- ▶ D_i The maximum acceptable travel time from demand $i \in I$ (the cover time).

$N_i = \{j \in J : d_{ij} \leq D_i\}$: Set of all candidate locations that can cover demand i .

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- ▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.

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- ▶ w_i The demand point at $i \in I$.
- ▶ p The number of candidate locations to be established.
- ▶ $x_j \in \{0, 1\}$ 1, if a facility is located at candidate location $j \in J$; 0 otherwise.
- ▶ $z_i \in \{0, 1\}$ 1, if demand point $i \in I$ is covered; 0 otherwise.

Maximal covering [Ahmadi-Javid et al., 2017]

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▶ $y_{ij} \in \{0, 1\}$ 1, if demand point i is assigned to a facility at candidate location $j \in N_i$; 0 otherwise.

▶ $L \geq 0$ L is an auxiliary variable (not a decision variable) that is used to compute the maximum distance.

min. L

$$\text{s.t.: } \sum_{j \in N_i} y_{ij} = 1 \quad \forall i \in I$$

$$\sum_{j \in J} x_j = p$$

$$\sum_{j \in N_i} w_i d_{ij} y_{ij} \leq L \quad \forall i \in I$$

$$y_{ij} \leq x_j \quad \forall i \in I, j \in N_i$$

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We provide a more detailed description of location problems **in health care**. See **Chapter 23**[Contreras and Ortiz-Astorquiza, 2019]

Consider a classical P-median to decide where to locate a set of primary care facilities, such as community clinics or family centers, or hospitals. E.g.:

I : The set of districts where population resides.

J : The set of potential locations for health care facilities.

- ▶ D_i : Population of district $i \in I$;
- ▶ C_{ij} : Distance from district $i \in I$ to potential location $j \in J$;
- ▶ P : Number of facilities to install;
- ▶ $x_{ij} \in \{0, 1\}$: Allocation of population of district $i \in I$ to location j
- ▶ $y_j \in \{0, 1\}$: If facility is open (or not) at location $j \in J$.

$$\min. \sum_{i \in I} \sum_{j \in J} D_i C_{ij} x_{ij} \quad (23.3)$$

$$\text{s.t.: } \sum_{j \in J} x_{ij} = 1 \quad \forall i \in I \quad (23.4)$$

$$x_{ij} \leq y_j \quad \forall i \in I, j \in J \quad (23.5)$$

$$\sum_{j \in J} y_j = P \quad (23.6)$$

$$y_j \in \{0, 1\} \quad \forall j \in J \quad (23.7)$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in I, j \in J \quad (23.8)$$

Objective Funtion

Objective Funtion: Healthcare facility location problems are **inherently multi-objective** because there are different stakeholders, such as patients, health providers and community. Facilities are majorly public.

- ▶ **Patients:** Travel time/cost, service quality or cost, waiting time;
- ▶ **Providers:** Setup and operating costs, staff availability, travel time/cost;
- ▶ **Community:** Equity in access among different districts.

Goals:

- ▶ Minimize access cost for health care patients;
- ▶ Minimize facilities costs;
- ▶ Maximize equity in access to a healthcare facility (covered demand).

Modelling capacity

Resource-based view of facilities: Health workforce, beds, equipment. etc. E.g.:

R : The set of resource types.

S : The set of service types

$R_s \subseteq R$: Resource required to serve demand for service $s \in S$

- ▶ D_{is} : Population of district $i \in I$ requiring service s ;
- ▶ K_{sr} : Amount of resource r to serve a patient requiring service s ;
- ▶ $x_{ijs} \in \{0, 1\}$: Allocation of population of district $i \in I$ for service s in location j
- ▶ q_{js} : Decisions on capacity for service s in location j (pop.)
- ▶ w_{rj} : Decisions on the amount of resource r in location j

Constraints:

$$\sum_{s \in S: r \in R_s} K_{sr} q_{js} \leq w_{rj} \quad \forall j \in J, r \in R \quad (23.10)$$

$$\sum_{i \in I} D_{is} x_{ijs} \leq q_{js} \quad \forall j \in J, s \in S \quad (23.11)$$

$$\sum_{j \in J} x_{ijs} = 1 \quad \forall i \in I, s \in S \quad (23.12)$$

Modelling allocation

Capacity limited system: In addition to the traditional single assignment assumption, which is the most common, there is multiple assignment in the case of limited capacity. However, it does not account for patient preference. E.g.:

J : Destination location.

S : The set of service types

- ▶ $x_{ijs} \geq 0$: The number of patients from district $i \in I$ assigned to service s in location j

Constraint:

$$\sum_{i \in I} x_{ijs} = D_{is} \quad \forall j \in J, s \in S \quad (23.14)$$

Closest assignment: It accounts for patient preference for the closest health unit. E.g.:

- ▶ $x_{ijs} \in \{0, 1\}$: Allocation of population of district $i \in I$ for service s in location j

Constraint:

$$\sum_{k \in J: C_{ik} > C_{ij}} \sum_{s \in S} x_{iks} + y_j \leq 1 \quad \forall i \in I, j \in J \quad (23.15)$$

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